

Lecture 8.

$$\xi : S \rightarrow \mathbb{R}.$$

discrete random variable. ξ .

Def (probability mass function / pmf).

$$\begin{aligned} P_{\xi}(x) &:= \mathbb{P}\{\xi = x\} \\ &= \mathbb{P}(\{w \in S : \xi(w) = x\}). \end{aligned}$$

e.g. (Toss a coin 3 times).

$$S = \{HHH, HHT, HTH, \dots\}.$$

$$\xi = \# \text{ heads. } \xi \text{ takes } 0, 1, 2, 3.$$

$$\{\xi = 0\} = \underline{\underline{\{TTT\}}}.$$

$$\{\xi = 1\} = \underline{\underline{\{HTT, THT, TTH\}}}.$$

$$\{\xi = 2\} = \{TTH, HTH, HHT\}.$$

$$\{\xi = 3\} = \{HHH\}.$$

x	0	1	2	3
$P_{\xi}(x)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

pmf for ξ .

e.g. (Shoot a target)

each shoot : β . repeat till hit.

$\sum$ = number of shots.

$\sum = 1, 2, \dots$

$$P_{\sum}(k) = \mathbb{P}\{\sum = k\} = (1-\beta)^{k-1} \cdot \beta$$

eg (Indicator random variable)

$A \in \mathcal{F}$. define

$$1_{\{A\}}(w) := \begin{cases} 1 & , w \in A \\ 0 & , w \notin A \end{cases}$$

$$\mathbb{P}\{1_{\{A\}} = 1\} = \mathbb{P}(A).$$

$$\mathbb{P}\{1_{\{A\}} = 0\} = \mathbb{P}(A^c).$$

def (pmf) $P_{\sum}(x) = \mathbb{P}\{\sum = x\}.$

def (Cumulative distribution function /cdf),

$$F_{\sum}(x) := \mathbb{P}\{\sum \leq x\}.$$

pmf $\Rightarrow$ cdf:

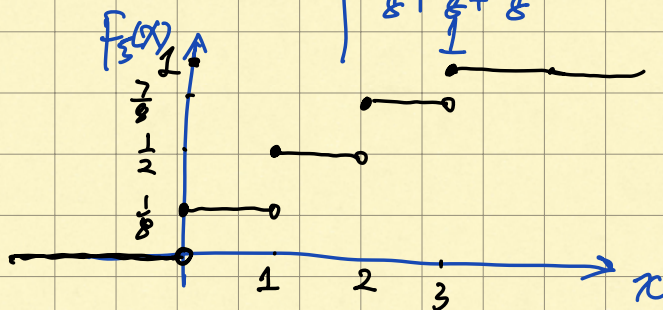
$$F_Z(x) = \mathbb{P}\{Z \leq x\}$$

$$= \sum_{y \leq x} \mathbb{P}\{Z=y\} = \sum_{y \leq x} P_Z(y).$$

x	0	1	2	3
$P_Z(x)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{4}$

$$F_Z(x) = \begin{cases} 0 & x < 0 \\ \frac{1}{8} & 0 \leq x < 1 \\ \frac{1}{8} + \frac{3}{8} & 1 \leq x < 2 \\ \frac{1}{8} + \frac{3}{8} + \frac{3}{8} & 2 \leq x < 3 \\ 1 & x \geq 3 \end{cases}$$

$$\begin{aligned} & x < 0. \\ & 0 \leq x < 1. \\ & 1 \leq x < 2. \\ & 2 \leq x < 3. \\ & x \geq 3. \end{aligned}$$



$$\text{cdf} \Leftrightarrow \begin{cases} \text{① non decreasing.} \\ \text{② right continuous} \\ \text{③ } F(+\infty) = 1 \\ F(-\infty) = 0. \end{cases}$$

$$\text{cdf} \Rightarrow \text{pmf}.$$

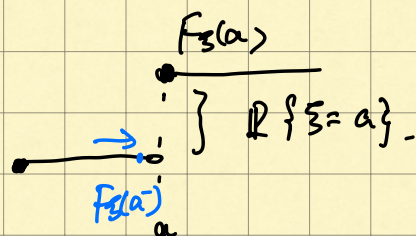
$$\bullet \quad \forall a < b.$$

$$\mathbb{P}\{a < Z \leq b\} = F_Z(b) - F_Z(a).$$

$$\bullet \quad \mathbb{P}\{Z = a\} = F_Z(a) - F_Z(a^-).$$

$$F_Z(a^-) = \lim_{x \rightarrow a^-} F_Z(x)$$

$\xrightarrow{x \rightarrow a^-}$ from left.



e.g. cdf pmf.

(Three dice) let M be the largest face.
pmf for M .

M can take 1, 2, 3, 4, 5, 6.

$$\cdot \mathbb{P}\{M=1\} = \left(\frac{1}{6}\right)^2$$

$$\cdot \mathbb{P}\{M=2\} =$$

$$\mathbb{P}\{M=4\} =$$

1, 2, 4
1, 1, 4
1, 2, 3
1, 4, 4

$$\cdot \mathbb{P}\{M=6\} = \left(\frac{1}{6}\right)^3$$

cdf $\Rightarrow$ pmf.

$$F_3(k) = \mathbb{P}\{M \leq k\} = \left(\frac{k}{6}\right)^3 \quad \checkmark \quad k=1, \dots, 6$$

$$P_3(k) = \mathbb{P}\{M=k\} = F_3(k) - F_3(k-1).$$

$$= \frac{k^3 - (k-1)^3}{6^3}$$

Expected Value of discrete R. V.

e.g. 100 students | 5 students | 0 course

10	1
30	2
30	3
25	4

On average, how many courses does a student take?

$$\frac{\text{Total Courses}}{\text{Total Students}} = \frac{5 \cdot 0 + 10 \cdot 1 + 30 \cdot 2 + 30 \cdot 3 + 25 \cdot 4}{100}$$

$$= \underline{0.05} \cdot \boxed{0} + \underline{0.1} \cdot \boxed{1} + \underline{0.3} \cdot \boxed{2} + \underline{0.3} \cdot \boxed{3} + \underline{0.25} \cdot \boxed{4}$$

$$= 2.6.$$

Def. (Expected Value) ξ takes $\{x_1, x_2, \dots\}$.

$$p_i = \mathbb{P}\{\xi = x_i\}.$$

$$E(\xi) = \mu_\xi := \sum_{i=1}^{\infty} x_i p_i.$$

$$E(\xi) = \sum_{k=1}^{\infty} k \mathbb{P}\{\xi = k\}$$

$$= \sum_{k=1}^{\infty} k \cdot (1-\beta)^{k-1} \cdot \beta$$

$$= \beta \left[\sum_{k=1}^{\infty} k (1-\beta)^{k-1} \right]$$

$$= \beta \cdot \frac{1}{\beta^2} = \frac{1}{\beta}.$$

$$E(S) = \infty.$$

insurance loss S = money of a claim.

$P\{S = \$1m\}$ is very small.

$$\sum \underline{\$k} \times P\{S = k\}$$

$$P\{S = k\} = \frac{a}{k^2}, \quad k=1, 2, 3, \dots$$

$$E(S) = \sum k \cdot \frac{a}{k^2} = a \cdot \left[\sum_{k=1}^{\infty} \frac{1}{k} \right] = \infty.$$

Functions of R.V.

$$S : \Omega \rightarrow \mathbb{R}.$$

$$h : \mathbb{R} \rightarrow \mathbb{R}.$$

$$h(x) = x^2$$

$$h(x) = \frac{1}{1+x^2}$$

$h \circ S : \Omega \rightarrow \mathbb{R}$ is a random variable.

S	1	2	3	4	...
Total cost	$5 + \log(1)$	$10 + \log(2)$	...	...	
$P_S(k)$	β	$(1-\beta)\beta$	$(1-\beta)^2\beta$	$(1-\beta)^3\beta$	...

bullet cost \$5

maintenance cost $\log(S)$

Total cost

$$S = k$$

$$\text{Total cost} = 5 \cdot k + \log(k).$$

$$h(\xi) := 5\xi + \log(\xi).$$

$$\mathbb{E}(h(\xi)) = \sum_{k=1}^{\infty} h(k) \cdot p_{\xi}(k).$$

ξ take $\{x_1, x_2, \dots\}$ $h: \mathbb{R} \rightarrow \mathbb{R}$.
 then $h(\xi)$ is a r.v.

$$\mathbb{E}(h(\xi)) = \sum_{i=1}^{\infty} \underline{h(x_i)} \mathbb{P}\{\xi = x_i\}.$$

if $h(x) = ax + b$.

$$\begin{aligned} \mathbb{E}(h(\xi)) &= \sum_{i=1}^{\infty} (ax_i + b) \mathbb{P}\{\xi = x_i\} \\ &= \underline{a \sum x_i \mathbb{P}\{\xi = x_i\}} + \underline{b \sum \mathbb{P}\{\xi = x_i\}} \\ &= a \mathbb{E}(\xi) + b. \end{aligned}$$

$$\Rightarrow \mathbb{E}(a\xi + b) = a \mathbb{E}(\xi) + b.$$

e.g. (Runs in a shuffled deck).

26 red, 26 black cards. is shuffled.

A run is max block of same color cards.

RR BR RR BB.
 has four runs.

$$\mathbb{E}(\text{runs}) = ?$$

$C_i = \{ \text{cards } i \text{ and } i+1 \text{ differ in color} \}$

1_{C_i}

$$\text{runs} = 1 + \sum_{i=1}^{51} 1_{C_i}$$

$$E(\text{runs}) = 1 + \sum_{i=1}^{51} E(1_{C_i})$$

$$= 1 + \sum_{i=1}^{51} P(C_i)$$

$$= 1 + \sum_{i=1}^{51} \frac{26}{52} \cdot \frac{26}{51} \cdot 2$$